MAT 3007 – Optimization

KKT Condition

Lecture 20

Instructor: Junchi Yang

CUHK-Shenzhen

Recap

The Lagrangian

$\min_x f(x)$

s.t. $g_i(x) \le 0$, $i = 1, \dots, m$

$h_j(x) = 0$, $j = 1, \dots, p$

We define the Lagrangian of this problem by:

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) := f(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i g_i(\boldsymbol{x}) + \sum_{j=1}^{p} \mu_j h_j(\boldsymbol{x})$$

where we associate each constraint with a Lagrangian multiplier:

$$\begin{aligned} g_i(\boldsymbol{x}) \le 0 \quad &\to \quad \lambda_i \ge 0 \\ h_j(\boldsymbol{x}) = 0 \quad &\to \quad \mu_j \end{aligned}$$

Lagrangian Multipliers:

- The multipliers for inequality constraints $g_i(\boldsymbol{x}) \leq 0$ are non-negative: $\lambda_i \geq 0$.
- The multipliers for the equality constraints are free: $\mu_j \in \mathbb{R}$.
- The condition $\lambda_i \geq 0$, μ_i free" are called dual feasibility.

Lagrangian Multipliers:

- The multipliers for inequality constraints $g_i(\boldsymbol{x}) \leq 0$ are non-negative: $\lambda_i \geq 0$.
- The multipliers for the equality constraints are free: $\mu_j \in \mathbb{R}$.
- The condition "$\lambda_i \geq 0$, μ_i free" are called dual feasibility.

Meaning of the Lagrangian:

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) := f(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i g_i(\boldsymbol{x}) + \sum_{j=1}^{p} \mu_j h_j(\boldsymbol{x})$$

- $\rightsquigarrow$ We move all the constraints to the objective and interpret them as reward/penalty functions.
- We want $g_i(\boldsymbol{x}) \leq 0$: setting $\lambda_i \geq 0$, we reduce the Lagrangian when the constraint holds and penalize/increase it when the constraint does not hold.

Theorem: KKT Conditions

If $\boldsymbol{x}$ is a local minimizer and if a constraint qualification holds, then there exist $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}$ such that:

1. **Main Condition**

$$\nabla_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \nabla f(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i \nabla g_i(\boldsymbol{x}) + \sum_{j=1}^{p} \mu_j \nabla h_j(\boldsymbol{x}) = \mathbf{0}.$$

2. **Dual Feasibility**

$$\lambda_i \geq 0 \quad i = 1, \ldots, m.$$

3. **Complementarity**

$$\lambda_i \cdot g_i(\boldsymbol{x}) = 0 \quad \forall\ i = 1, \ldots, m.$$

4. **Primal Feasibility**

$$g_i(\boldsymbol{x}) \leq 0, \quad h_j(\boldsymbol{x}) = 0 \quad \forall\ i, \quad \forall\ j.$$

Constraint Qualifications

We require the collection of gradients

$$\{\nabla g_i(\boldsymbol{x}) : i \in \mathcal{A}(\boldsymbol{x})\} \cup \{\nabla h_j(\boldsymbol{x}) : j = 1, \ldots, p\} \qquad (\star)$$

to be linearly independent.

- This condition is a constraint qualification (CQ) called the Linear Independence Constraint Qualification (LICQ).

Constraint Qualifications

We require the collection of gradients

$$\{\nabla g_i(\boldsymbol{x}) : i \in \mathcal{A}(\boldsymbol{x})\} \cup \{\nabla h_j(\boldsymbol{x}) : j = 1, \ldots, p\} \qquad (\star)$$

to be linearly independent.

- This condition is a constraint qualification (CQ) called the Linear Independence Constraint Qualification (LICQ).
- A feasible point $\boldsymbol{x}$ satisfying LICQ is called regular.
- LICQ ensures that the Karush-Kuhn-Tucker (KKT) conditions are necessary for optimality.

Constraint Qualifications

We require the collection of gradients

$$\{\nabla g_i(\boldsymbol{x}) : i \in \mathcal{A}(\boldsymbol{x})\} \cup \{\nabla h_j(\boldsymbol{x}) : j = 1, \ldots, p\} \qquad (\star)$$

to be linearly independent.

- This condition is a constraint qualification (CQ) called the Linear Independence Constraint Qualification (LICQ).
- A feasible point $\boldsymbol{x}$ satisfying LICQ is called regular.
- LICQ ensures that the Karush-Kuhn-Tucker (KKT) conditions are necessary for optimality.
- For linear constraints: No additional condition is required for the KKT conditions to be necessary.

Failure of KKT Conditions in the Absence of CQ

$$\begin{aligned} \text{minimize}_{x_1,x_2} \quad & x_1 \\ \text{subject to} \quad & (x_1-1)^2+(x_2-1)^2 \le 1, \\ & (x_1-1)^2+(x_2+1)^2 \le 1. \end{aligned}$$

- Since there is only one feasible solution (i.e., $(x_1, x_2) = (1, 0)$), it is automatically optimal.

Failure of KKT Conditions in the Absence of CQ

$$\begin{aligned} \text{minimize}_{x_1,x_2} \quad & x_1 \\ \text{subject to} \quad & (x_1-1)^2+(x_2-1)^2 \le 1, \to \lambda_1 \\ & (x_1-1)^2+(x_2+1)^2 \le 1. \to \lambda_2 \end{aligned}$$

- Since there is only one feasible solution (i.e., $(x_1, x_2) = (1, 0)$), it is automatically optimal.
- However, this point does not satisfy the KKT conditions.

$L(x,\lambda_1,\lambda_2) = x_1 + \lambda_1\left[(x_1-1)^2+(x_2-1)^2-1\right] + \lambda_2\left[(x_1-1)^2+(x_2+1)^2-1\right]$.

main condition : $\nabla_x L(x,\lambda_1,\lambda_2) = \begin{bmatrix} -1+\lambda_1(2x_1-2)+\lambda_2(2x_1-2) \\ \lambda_1(2x_2-2)+\lambda_2(2x_2+2) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

$\Rightarrow \begin{cases} \lambda_1(x_1-1)+\lambda_2(x_1-1) = \frac{1}{2} \\ \lambda_1(x_2-1)+\lambda_2(x_2+1) = 0 \end{cases}$ $\to$ this is not satisfied with $x_1=1$

so this point is NOT a KKT point.

$$\nabla g_1(x) = \begin{bmatrix} 2x_1 - 2 \\ 2x_2 - 2 \end{bmatrix} \qquad \nabla g_2(x) = \begin{bmatrix} 2x_1 - 2 \\ 2x_2 + 2 \end{bmatrix}$$

At point $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, both constraints are active. and

$$\nabla g_1(x) = \begin{bmatrix} 0 \\ -2 \end{bmatrix} \quad \text{and} \quad \nabla g_2(x) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

they are linearly dependent,

$\longrightarrow$ LICQ does not hold at this point.

Failure of KKT Conditions in the Absence of CQ

$$\begin{aligned} \text{minimize}_{x_1,x_2} \quad & x_1 \\ \text{subject to} \quad & (x_1-1)^2+(x_2-1)^2 \le 1, \\ & (x_1-1)^2+(x_2+1)^2 \le 1. \end{aligned}$$

- Since there is only one feasible solution (i.e., $(x_1, x_2) = (1, 0)$), it is automatically optimal.
- However, this point does not satisfy the KKT conditions.
- This example says if a local min does not hold certain CQ, the KKT conditions may not be necessary at this point.

Remarks on KKT Conditions

Remarks:

- KKT conditions is first-order necessary conditions (FONC) for general constrained optimization problems.

Remarks on KKT Conditions

Remarks:

- KKT conditions is first-order necessary conditions (FONC) for general constrained optimization problems.
- A (feasible) point (often the primal and dual variables $(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ together) satisfying the KKT conditions is called a KKT point regardless whether it satisfies CQ or not. We may also call $\boldsymbol{x}$ a KKT point for simplicity.

Remarks on KKT Conditions

Remarks:

- KKT conditions is first-order necessary conditions (FONC) for general constrained optimization problems.
- A (feasible) point (often the primal and dual variables $(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ together) satisfying the KKT conditions is called a KKT point regardless whether it satisfies CQ or not. We may also call $\boldsymbol{x}$ a KKT point for simplicity.
- At a KKT point, the Lagrange multipliers (dual variables) $\lambda_i = 0, i \in I(\boldsymbol{x})$.

Because by complem. condition,

$\lambda_i \underbrace{g_i(x)}_{<0 \text{ for } i \in I(x)} = 0.$

Remarks on KKT Conditions

Remarks:

- KKT conditions is first-order necessary conditions (FONC) for general constrained optimization problems.
- A (feasible) point (often the primal and dual variables $(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ together) satisfying the KKT conditions is called a KKT point regardless whether it satisfies CQ or not. We may also call $\boldsymbol{x}$ a KKT point for simplicity.
- At a KKT point, the Lagrange multipliers (dual variables) $\lambda_i = 0, i \in I(\boldsymbol{x})$.
- If a local minimizer $\boldsymbol{x}$ satisfies LICQ, then the Lagrange multipliers (dual variables) $\boldsymbol{\lambda}, \boldsymbol{\mu}$ solved by the KKT conditions must be unique (why?).

main condition: $\nabla f(x) + \sum_i \lambda_i \nabla g_i(x) + \sum_j \mu_j \nabla h_j(x) = 0$

Note $\lambda_i = 0$ for $i \in I(x)$

plug it back to main condition:

$$\nabla f(x) + \sum_{i \in A(x)} \lambda_i \nabla g_i(x) + \sum_j \mu_j \nabla h_j(x) = 0$$

Because x is a KKT point, the linear system above must have a solution λ and μ. Also, by LICQ, we have $\{\nabla g_i(x) : i \in A(x)\} \cup \{\nabla h_j(x)\}$ are linearly independent, then λ and μ are unique. (by property of linear independence)

Strategy for Applying KKT Conditions

General Strategy:

- Check LICQ (if required).
- Derive KKT conditions.
- Start by discussing different easy cases via the complementarity conditions (set multiplier or constraints to 0), then find all KKT points.

Examples: Formulating and Using KKT Conditions

Example III: Cylinder Volume

We want to build a cylinder with the maximum volume, with its surface area no larger than C.

- Decision variables: r (the radius of the base) and h (height).

$$\begin{aligned} \text{maximize}_{r,h} \quad & \pi r^2 h \\ \text{subject to} \quad & 2\pi r^2 + 2\pi rh \leq C \\ & r, h \geq 0 \end{aligned}$$

Example III: Cylinder Volume

We want to build a cylinder with the maximum volume, with its surface area no larger than C.

- Decision variables: r (the radius of the base) and h (height).

$$\begin{aligned} \text{maximize}_{r,h} \quad & \pi r^2 h \\ \text{subject to} \quad & 2\pi r^2 + 2\pi rh \leq C \\ & r, h \geq 0 \end{aligned}$$

- Denote $\boldsymbol{x} = (r; h)$. The constraints are

$$\begin{aligned} g_1(\boldsymbol{x}) &= 2\pi r(r+h) - C \leq 0, \\ g_2(\boldsymbol{x}) &= -r \leq 0, \\ g_3(\boldsymbol{x}) &= -h \leq 0. \end{aligned}$$

Example III: Continued

Convert it to a minimization problem:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}=(r,h)} \quad & -\pi r^2 h \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \le 0, \quad i = 1, 2, 3. \end{aligned}$$

Introduce Lagrangian multiplier $\boldsymbol{\lambda} \ge \mathbf{0}$. The Lagrangian is:

$$L(r, h, \boldsymbol{\lambda}) = -\pi r^2 h + \boldsymbol{\lambda}^\top g(\boldsymbol{x}) \qquad \lambda = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix}$$

We have

$$\nabla_{\boldsymbol{x}} L(r, h, \boldsymbol{\lambda}) = \begin{pmatrix} -2\pi r h \\ -\pi r^2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 4\pi r + 2\pi h \\ 2\pi r \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \lambda_3 \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

$\nabla f(x) \qquad \nabla g_1(x) \qquad \nabla g_2(x) \qquad \nabla g_3(x)$

Example III — KKT Conditions

1. Main condition:

$$-2\pi rh + 4\pi r\lambda_1 + 2\pi h\lambda_1 - \lambda_2 = 0, \quad -\pi r^2 + 2\pi r\lambda_1 - \lambda_3 = 0$$

2. Dual feasibility: $\boldsymbol{\lambda} \geq \mathbf{0}$
3. Complementarity conditions:

$$\begin{aligned} \lambda_1 \cdot (2\pi r^2 + 2\pi rh - C) &= 0 \\ \lambda_2 \cdot r &= 0 \\ \lambda_3 \cdot h &= 0 \end{aligned}$$

4. Primal feasibility:

$$2\pi r^2 + 2\pi rh \leq C, \quad r \geq 0, \quad h \geq 0$$

Example III: Continued

Case 1: $r = 0$. Then, the main condition reduces to

$$2\pi h\lambda_1 - \lambda_2 = 0, \quad \lambda_3 = 0.$$

Since $r = 0$, we have $g_1(x) = -C < 0$. By the first complem. condition, we must have: $\lambda_1 = 0$. Therefore, $\boldsymbol{\lambda} = \mathbf{0}$.

Example III: Continued

Case 1: $r = 0$. Then, the main condition reduces to

$$2\pi h\lambda_1 - \lambda_2 = 0, \quad \lambda_3 = 0.$$

Since $r = 0$, we have $g_1(x) = -C < 0$. By the first complem. condition, we must have: $\lambda_1 = 0$. Therefore, $\boldsymbol{\lambda} = \mathbf{0}$.

- We plug in $r = 0$ and $\boldsymbol{\lambda} = \mathbf{0}$ back into KKT conditions. It is easy to conclude that all $(0; h)$ with $h \geq 0$ are KKT points.

Example III: Continued

Case 1: $r = 0$. Then, the main condition reduces to

$$2\pi h\lambda_1 - \lambda_2 = 0, \quad \lambda_3 = 0.$$

Since $r = 0$, we have $g_1(x) = -C < 0$. By the first complem. condition, we must have: $\lambda_1 = 0$. Therefore, $\boldsymbol{\lambda} = \mathbf{0}$.

- We plug in $r = 0$ and $\boldsymbol{\lambda} = \mathbf{0}$ back into KKT conditions. It is easy to conclude that all $(0; h)$ with $h \geq 0$ are KKT points.

Case 2: $r > 0$. By second complem. condition, we have $\lambda_2 = 0$.

Example III: Continued

Case 1: $r = 0$. Then, the main condition reduces to

$$2\pi h\lambda_1 - \lambda_2 = 0, \quad \lambda_3 = 0.$$

Since $r = 0$, we have $g_1(x) = -C < 0$. By the first complem. condition, we must have: $\lambda_1 = 0$. Therefore, $\boldsymbol{\lambda} = \mathbf{0}$.

- We plug in $r = 0$ and $\boldsymbol{\lambda} = \mathbf{0}$ back into KKT conditions. It is easy to conclude that all $(0; h)$ with $h \geq 0$ are KKT points.

Case 2: $r > 0$. By second complem. condition, we have $\lambda_2 = 0$.
Subcase 2.1: $\lambda_1 = 0$. By the second main condition, we have $\lambda_3 = -\pi r^2 < 0$, so there is no such KKT points.

Example III: Continued

Case 1: $r = 0$. Then, the main condition reduces to

$$2\pi h\lambda_1 - \lambda_2 = 0, \quad \lambda_3 = 0.$$

Since $r = 0$, we have $g_1(x) = -C < 0$. By the first complem. condition, we must have: $\lambda_1 = 0$. Therefore, $\boldsymbol{\lambda} = \mathbf{0}$.

- We plug in $r = 0$ and $\boldsymbol{\lambda} = \mathbf{0}$ back into KKT conditions. It is easy to conclude that all $(0; h)$ with $h \geq 0$ are KKT points.

Case 2: $r > 0$. By second complem. condition, we have $\lambda_2 = 0$.
Subcase 2.1: $\lambda_1 = 0$. By the second main condition, we have $\lambda_3 = -\pi r^2 < 0$, so there is no such KKT points.
Subcase 2.2: $\lambda_1 > 0$. We have $h \neq 0$ (otherwise the first main condition yields $4\pi r\lambda_1 = 0$). Also, by third complem. condition, we have $\lambda_3 = 0$.

Example III: Continued

Therefore, in Case 2, $\lambda_1 > 0$ and $\lambda_2 = \lambda_3 = 0$.

Example III: Continued

Therefore, in Case 2, $\lambda_1 > 0$ and $\lambda_2 = \lambda_3 = 0$. Then the second main condition implies $\lambda_1 = r/2$. Plug it back in the first main condition, we get $h = 2r$. Because $\lambda_1 > 0$, by the first complem. condition, we have $2\pi r^2 + 2\pi rh = C$.

Example III: Continued

Therefore, in Case 2, $\lambda_1 > 0$ and $\lambda_2 = \lambda_3 = 0$. Then the second main condition implies $\lambda_1 = r/2$. Plug it back in the first main condition, we get $h = 2r$. Because $\lambda_1 > 0$, by the first complem. condition, we have $2\pi r^2 + 2\pi rh = C$. Combined with $h = 2r$, we solve $r = \sqrt{C/6\pi}, h = 2\sqrt{C/6\pi}$. And $\lambda_1 = r/2 = \sqrt{C/24\pi}$.

Example III: Continued

Therefore, in Case 2, $\lambda_1 > 0$ and $\lambda_2 = \lambda_3 = 0$. Then the second main condition implies $\lambda_1 = r/2$. Plug it back in the first main condition, we get $h = 2r$. Because $\lambda_1 > 0$, by the first complem. condition, we have $2\pi r^2 + 2\pi rh = C$. Combined with $h = 2r$, we solve $r = \sqrt{C/6\pi}, h = 2\sqrt{C/6\pi}$. And $\lambda_1 = r/2 = \sqrt{C/24\pi}$.

To conclude, all KKT points for this problem are $\{(0; h) : h \geq 0\}$ and $(\sqrt{C/6\pi}; 2\sqrt{C/6\pi})$.

Example III: Continued

Question: Can a point that does not satisfy the KKT conditions potentially be a local minimizer?

Example III: Continued

Question: Can a point that does not satisfy the KKT conditions potentially be a local minimizer?

$\rightsquigarrow$ Check LICQ.

$$g_1(\boldsymbol{x}) = 2\pi r(r+h) - C \le 0,\ g_2(\boldsymbol{x}) = -r \le 0,\ g_3(\boldsymbol{x}) = -h \le 0.$$

$\nabla g_1(x) = \begin{bmatrix} 4\pi r + 2\pi h \\ 2\pi r \end{bmatrix}$ $\nabla g_2(x) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ $\nabla g_3(x) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$.

Let $\mathcal{A}(x)$ be the set of active index

<1> $\mathcal{A}(x) = \{1,2,3\} \Leftrightarrow r = h = 0$ and $-C = 0$. Impossible.

<2> $\mathcal{A}(x) = \{1,2\} \Leftrightarrow r = 0,\ h \neq 0$ and $-C = 0$. Impossible.

<3> $\mathcal{A}(x) = \{2,3\} \Leftrightarrow r = h = 0$ and $-C < 0$

$\nabla g_2(x) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ $\nabla g_3(x) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ linearly independent ✓

<4> $\mathcal{A}(x) = \{1,3\} \Leftrightarrow h = 0,\ r \neq 0$ and $2\pi r^2 - C = 0$

Continue...

$\nabla g_1(x) = \begin{bmatrix} 4\pi r \\ 2\pi r \end{bmatrix}$ and $\nabla g_3(x) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ linearly indep because $r \neq 0$ ✓

<5> $A(x) = \{1\} \Leftrightarrow 2\pi r(r+h) - c = 0$, $r > 0$, $h > 0$.

$\nabla g_1(x) = \begin{bmatrix} 4\pi r \\ 2\pi r \end{bmatrix} \neq 0$ ✓

<6> $A(x) = \{2\} \Leftrightarrow r = 0$, $-c < 0$ and $h > 0$

$\nabla g_2(x) = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \neq 0$ ✓

<7> $A(x) = \{3\} \Leftrightarrow h = 0$, $r \geq 0$ and $2\pi r^2 - c < 0$

$\nabla g_3(x) = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \neq 0$ ✓

<8> $A(x) = \emptyset$ ✓

Remark: For set $\{\nabla g_i(x), i \in A(x)\} \cup \{h_j(x), \forall j\}$.

① If it's empty, we consider LICQ holds trivially.

② If it has only one vector, we consider it linearly indep. if it's nonzero.

Example IV

Task: We want to build a box with a given volume of at least 64 cubic inches. We want to minimize the total amount of material used.

Example IV

Task: We want to build a box with a given volume of at least 64 cubic inches. We want to minimize the total amount of material used.

We can formulate the optimization problem as:

$$\begin{aligned} \text{minimize} \quad & 2xy + 2yz + 2xz \\ \text{s.t.} \quad & xyz \geq 64 \end{aligned}$$

Define $g(x, y, z) = 64 - xyz$ and let $\lambda \geq 0$ be the Lagrange multiplier.

Example IV: Continued

The KKT conditions say: $xyz \geq 64$, and

$$2\begin{pmatrix} y+z \\ x+z \\ x+y \end{pmatrix} = \lambda \begin{pmatrix} yz \\ xz \\ xy \end{pmatrix}, \quad \lambda \geq 0, \quad \lambda \cdot (xyz - 64) = 0$$

Case 1: $\lambda = 0$. Then we must have $x = y = z = 0$, however, this point does not satisfy the constraint.

Example IV: Continued

The KKT conditions say: $xyz \geq 64$, and

$$2\begin{pmatrix} y+z \\ x+z \\ x+y \end{pmatrix} = \lambda \begin{pmatrix} yz \\ xz \\ xy \end{pmatrix}, \quad \lambda \geq 0, \quad \lambda \cdot (xyz - 64) = 0$$

Case 1: $\lambda = 0$. Then we must have $x = y = z = 0$, however, this point does not satisfy the constraint.

Case 2: $\lambda > 0$. By complem. condition, $xyz = 64$. The by the main condition

$$\lambda = 2\left(\frac{1}{x} + \frac{1}{y}\right) = 2\left(\frac{1}{y} + \frac{1}{z}\right) = 2\left(\frac{1}{x} + \frac{1}{z}\right)$$

Thus, $x = y = z = 4$ is the only solution of the KKT conditions.

Example IV: Continued

The KKT conditions say: $\mathsf{xyz} \geq 64$, and

$$2\begin{pmatrix} y+z \\ x+z \\ x+y \end{pmatrix} = \lambda \begin{pmatrix} yz \\ xz \\ xy \end{pmatrix}, \quad \lambda \geq 0, \quad \lambda \cdot (xyz - 64) = 0$$

Case 1: $\lambda = 0$. Then we must have $x = y = z = 0$, however, this point does not satisfy the constraint.

Case 2: $\lambda > 0$. By complem. condition, $xyz = 64$. The by the main condition

$$\lambda = 2\left(\frac{1}{x} + \frac{1}{y}\right) = 2\left(\frac{1}{y} + \frac{1}{z}\right) = 2\left(\frac{1}{x} + \frac{1}{z}\right)$$

Thus, $x = y = z = 4$ is the only solution of the KKT conditions.

Question: Can a point that does not satisfy the KKT conditions potentially be a local minimizer?

check LICQ.

$$g(x,y,z) = 64 - xyz$$

① $A=\{1\}$. $\nabla g(x,y,z) = \begin{bmatrix} -yz \\ -xz \\ -xy \end{bmatrix} \neq 0$ as long as $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq 0$

$x=y=z=0$ is not a feasible point

② $A=\phi$ ✓

$\Rightarrow$ LICQ is satified for any feasible point.

Visualization of the KKT Conditions

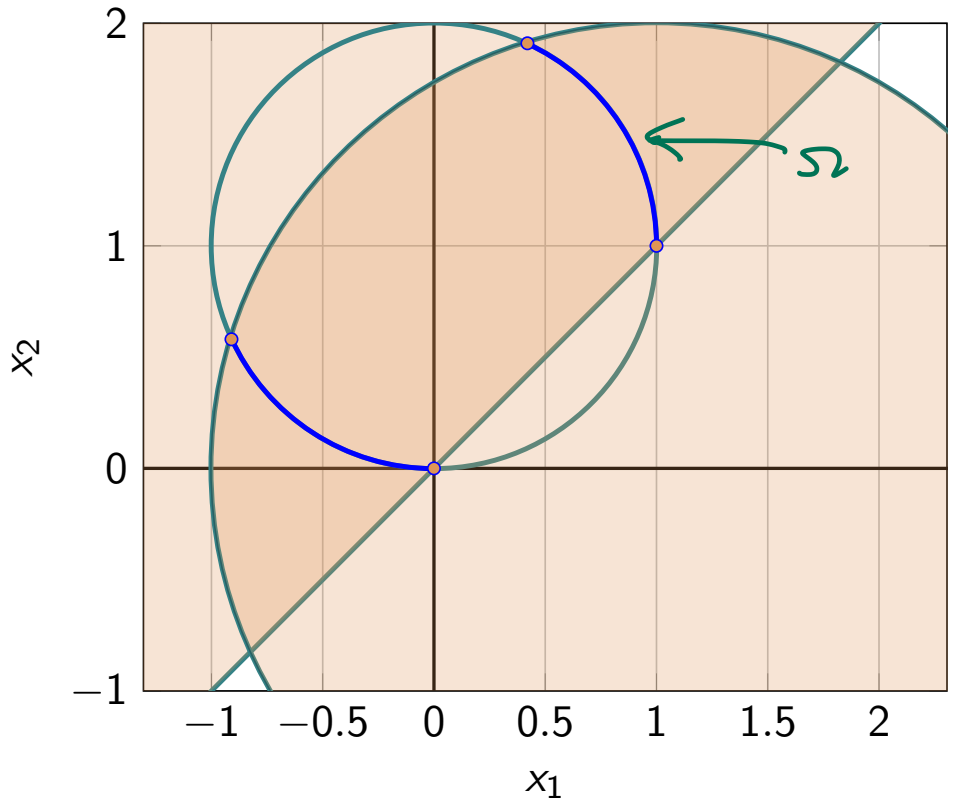
Example

Consider the problem

$$\min_{\boldsymbol{x}} \; x_1^3 x_2 - 2x_1^2 + 3x_2 \quad \text{s.t.} \quad g(\boldsymbol{x}) \leq \mathbf{0}, \quad h(\boldsymbol{x}) = 0,$$

where $g_1(\boldsymbol{x}) = (x_1 - 1)^2 + x_2^2 - 4$

$$g_2(\boldsymbol{x}) = x_1 - x_2, \quad h(\boldsymbol{x}) = x_1^2 + (x_2 - 1)^2 - 1.$$

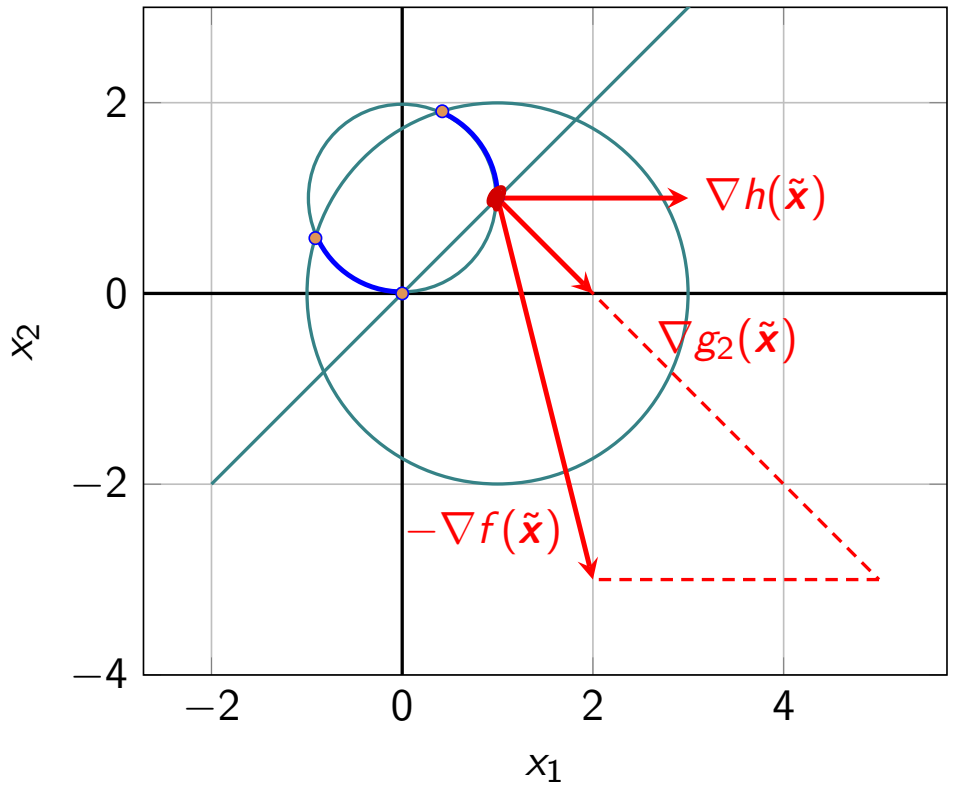


KKT-Conditions: Visualization

Consider this problem at $\tilde{\boldsymbol{x}} = (1,1)^\top$. We have $\mathcal{A}(\tilde{\boldsymbol{x}}) = \{2\}$ and

$$\nabla f(\tilde{\boldsymbol{x}}) = (-1,4)^\top,\ \nabla g_2(\tilde{\boldsymbol{x}}) = (1,-1)^\top,\ \nabla h(\tilde{\boldsymbol{x}}) = (2,0)^\top$$

The KKT-conditions mean: $-\nabla f(\tilde{\boldsymbol{x}}) = \nabla g_2(\tilde{\boldsymbol{x}})\lambda_2 + \nabla h(\tilde{\boldsymbol{x}})\mu$ for some $\lambda_2 \geq 0$ and μ:



Convexity

Towards Global Optimality

So far we have been discussing local minimizers:

- When is a local minimizer also a global minimizer?
- We present a class of optimization problems that guarantees this property $\rightsquigarrow$ convex optimization.

Definition: Convex Set

A set $\Omega \subseteq \mathbb{R}^n$ is convex if for any $\boldsymbol{x}$, $\boldsymbol{y} \in \Omega$, and any $\lambda \in [0, 1]$, $\lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y} \in \Omega$.

Definition: Convex Set

A set $\Omega \subseteq \mathbb{R}^n$ is convex if for any $\boldsymbol{x}$, $\boldsymbol{y} \in \Omega$, and any $\lambda \in [0, 1]$, $\lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y} \in \Omega$.

Examples:

- Let $\boldsymbol{a} \in \mathbb{R}^n$ and $b \in \mathbb{R}$ be given. The half space

$$H := \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{a}^\top \boldsymbol{x} \leq b\}$$

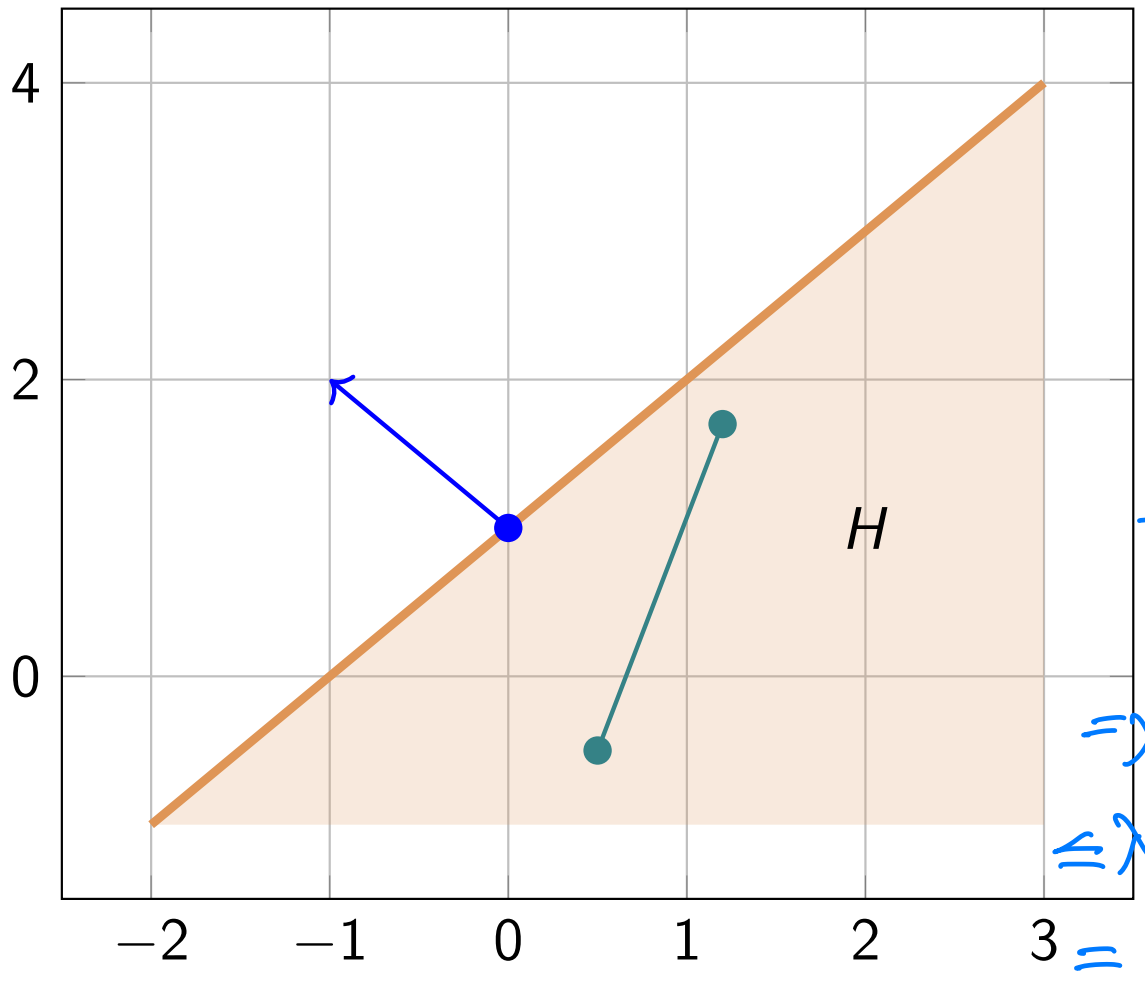
  is a convex set.

- Let $\boldsymbol{a} \in \mathbb{R}^n$ and $r > 0$ be given. The closed ball

$$\bar{B}_r(\boldsymbol{a}) := \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x} - \boldsymbol{a}\| \leq r\}$$

  is a convex set.

Illustration: Half Space



$\{x : a^T x \le b\}$

Why convex?

If x, y satisfy $a^T x \le b$, $a^T y \le b$, then

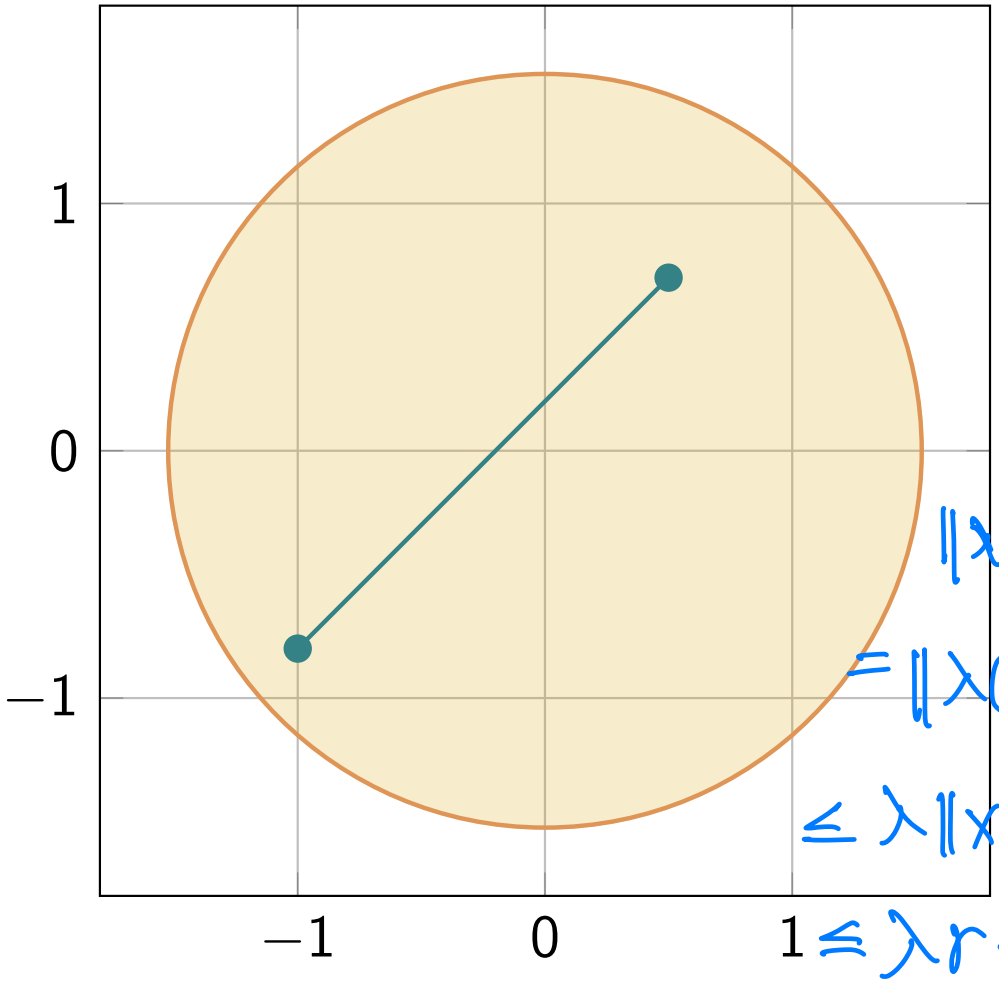
$a^T[\lambda x + (1-\lambda) y]$

$= \lambda a^T x + (1-\lambda) a^T y$

$\le \lambda b + (1-\lambda) b$

$= b$

Illustration: Ball



$\{x : \|x-a\| \le r\}$

Why convex ?

If $\|x-a\| \le r, \|y-a\| \le r$,

then

$\|\lambda x + (1-\lambda) y - a\|$

$= \|\lambda(x-a) + (1-\lambda)(y-a)\|$

$\le \lambda \|x-a\| + (1-\lambda)\|y-a\|$

$\le \lambda r + (1-\lambda) r$

$= r$

Definition: Convex Function

A function f on a convex set Ω is said to be convex if for every $\boldsymbol{x}, \boldsymbol{y} \in \Omega$ and any $0 \leq \lambda \leq 1$,

$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \leq \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}).$$

Convex Functions

Definition: Convex Function

A function f on a convex set Ω is said to be convex if for every $\boldsymbol{x}, \boldsymbol{y} \in \Omega$ and any $0 \leq \lambda \leq 1$,

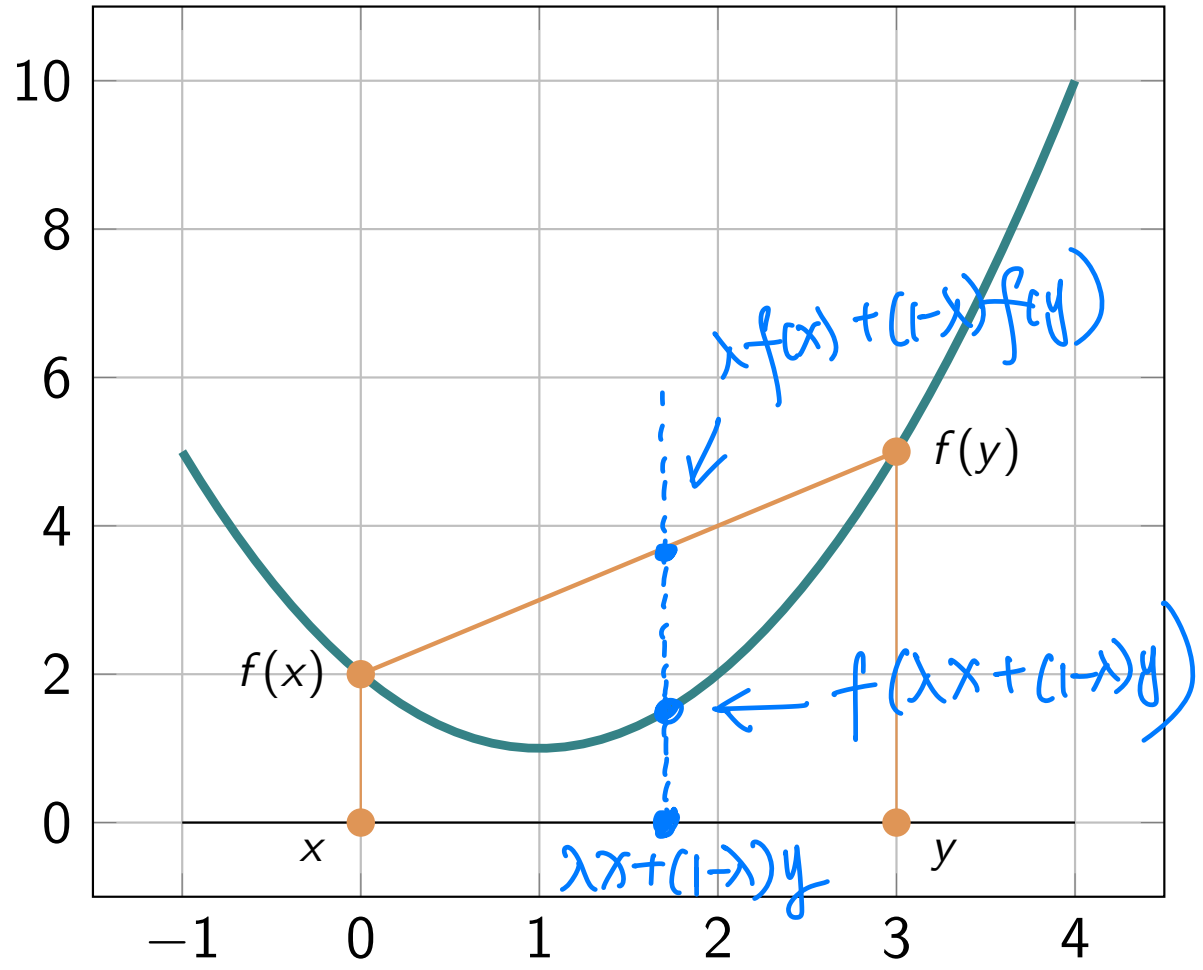
$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \leq \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}).$$

Definition: Concave Function

We call f a concave function if and only if $-f$ is convex, i.e., for any $\boldsymbol{x}, \boldsymbol{y} \in \Omega$ and $0 \leq \lambda \leq 1$,

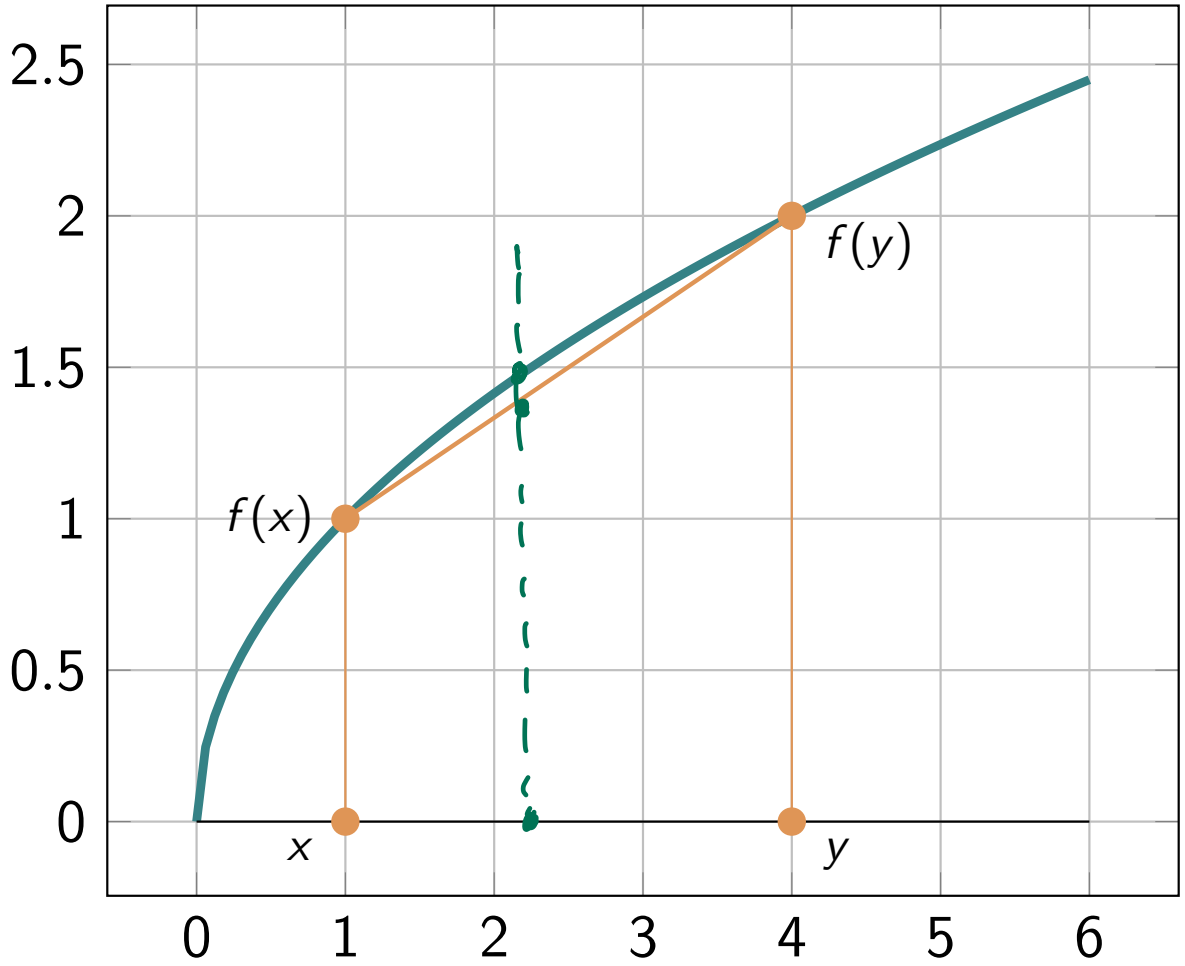
$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \geq \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}).$$

Illustration of Convex Functions

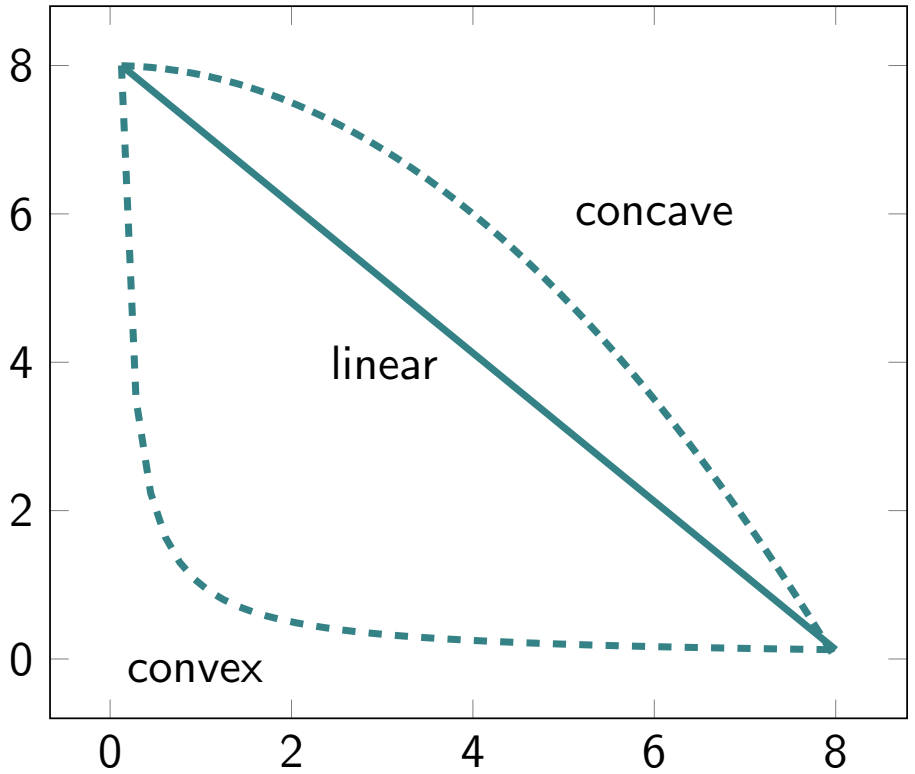


$$f(\lambda x+(1-\lambda)y) \le \lambda f(x) + (1-\lambda)f(y)$$

Illustration of Concave Functions



Convex Functions: Examples



Some examples of convex functions:

- $f(x) = x$, $f(x) = x^2$, $f(x) = e^x$, $f(x) = |x|$.

Some examples of concave functions:

- $f(x) = x$, $f(x) = \sqrt{x}$, $f(x) = \log x$

Examples: ① $f(x)=x$

$$\lambda f(x)+(1-\lambda)f(y)=\lambda x+(1-\lambda)y=f(\lambda x+(1-\lambda)y)$$ □

② $f(x)=x^2$.

$$\lambda f(x)+(1-\lambda)f(y)=\lambda x^2+(1-\lambda)y^2 \qquad <1>$$

$$f(\lambda x+(1-\lambda)y)=\lambda^2x^2+(1-\lambda)^2y^2+2\lambda(1-\lambda)xy. \qquad <2>$$

$$<1>-<2>=(\lambda-\lambda^2)x^2+[(1-\lambda)-(1-\lambda)^2]y^2-2\lambda(1-\lambda)xy$$

$$=(\lambda-\lambda^2)x^2+(\lambda-\lambda^2)y^2-2\lambda(1-\lambda)xy$$

$$=\underbrace{\lambda(1-\lambda)}_{\geq 0}\underbrace{[x^2+y^2-2xy]}_{=(x-y)^2\geq 0}\geq 0$$ □

Convex Calculus

Convexity via Derivatives

Theorem: Convexity via Hessian

Let f be twice cont. differentiable and let $\Omega \subseteq \mathbb{R}^n$ be convex and open. Then f is convex on Ω if and only if its Hessian matrix is positive semidefinite, i.e.,

$$\boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x}) \boldsymbol{d} \geq 0 \quad \forall\, \boldsymbol{d} \in \mathbb{R}^n, \quad \forall\, \boldsymbol{x} \in \Omega.$$

Theorem: Convexity via Hessian

Let f be twice cont. differentiable and let $\Omega \subseteq \mathbb{R}^n$ be convex and open. Then f is convex on Ω if and only if its Hessian matrix is positive semidefinite, i.e.,

$$\boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x}) \boldsymbol{d} \geq 0 \quad \forall\, \boldsymbol{d} \in \mathbb{R}^n, \quad \forall\, \boldsymbol{x} \in \Omega.$$

- In $\mathbb{R}$, this means that the second-order derivative is non-negative.
- **Examples**: check whether $x \log x$, $\|\boldsymbol{x}\|^2$ are convex?

① $f(x) = x \log x$

$f'(x) = 1 + \log x$

$f''(x) = \frac{1}{x} > 0$ for $x \in (0, \infty)$

② $f(x) = \|x\|^2$

$\nabla f(x) = 2x$

$\nabla^2 f(x) = 2I$

Theorem: Concavity via Hessian

Let f be twice cont. differentiable and let $\Omega \subseteq \mathbb{R}^n$ be convex and open. Then f is concave on Ω if and only if its Hessian matrix is negative semidefinite, i.e.,

$$\boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x}) \boldsymbol{d} \leq 0 \quad \forall\, \boldsymbol{d} \in \mathbb{R}^n, \quad \forall\, \boldsymbol{x} \in \Omega.$$

- In $\mathbb{R}$, this means that the second-order derivative is non-positive.
- **Examples:** $\sqrt{x}$, $\log x$.

Lemma: Sum Rule

If $a_1, \ldots, a_m \geq 0$, and $f_1, \ldots, f_m$ are convex (concave) functions, then $a_1 f_1 + \cdots + a_m f_m$ is a convex (concave) function.

- **Examples:** $x_1^2 + x_2^2$, $e^x + |x|$.

To prove it: <1> check: if $a \geq 0$, f is convex then af is still convex

$$a\lambda f(x) + a(1-\lambda) f(y) = a[\lambda f(x) + (1-\lambda) f(y)]$$
$$\geq a f(\lambda x + (1-\lambda) y)$$

<2> check: if f and g are convex, then $f+g$ is convex.

Lemma: Sum Rule

If $a_1, \ldots, a_m \geq 0$, and $f_1, \ldots, f_m$ are convex (concave) functions, then $a_1 f_1 + \cdots + a_m f_m$ is a convex (concave) function.

- **Examples:** $x_1^2 + x_2^2$, $e^x + |x|$.

Lemma: Composition with Linear Functions

If $f : \mathbb{R}^m \to \mathbb{R}$ is convex (concave) and $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, $\boldsymbol{b} \in \mathbb{R}^m$ are given, then $g : \mathbb{R}^n \to \mathbb{R}$, $g(\boldsymbol{x}) := f(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b})$, is convex (concave).

- **Examples:** e^{2x+3}, $(x_1 - x_2)^2 + (x_2 + x_3)^2$, $\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|$, or $\log(-2x_1 + 3x_2 + 5)$ (concave), ...

Today's lecture

- Examples for KKT Conditions
- Convex Function

Next lecture

- More about Convexity

Questions?